

CDP & LLDP — Complete Notes

1. Layer 2 Discovery Protocols

CDP and LLDP

CDP (Cisco Discovery Protocol) and **LLDP (Link Layer Discovery Protocol)** are **Layer 2 neighbor-discovery protocols**.

Their purpose is to allow a device to:

- Discover directly connected devices.
- Learn information about those devices.
- Share information about itself with its neighbor.

Typical information includes:

- Device/host name
- IP address
- Device type
- Local interface
- Neighbor's interface
- Device capabilities
- Software/IOS information
- Platform/model

Important

They are **not routing protocols**.

CDP / LLDP



Discover connected devices

OSPF



Discover routing information



Build routing table

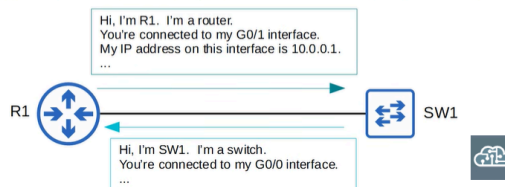
Image placement

📌 Place Image 1 here — *Layer 2 Discovery Protocols overview.*



Layer 2 Discovery Protocols

- Layer 2 discovery protocols such as CDP and LLDP share information with and discover information about neighboring (connected) devices.
- The shared information includes host name, IP address, device type, etc.
- CDP is a Cisco proprietary protocol.
- LLDP is an industry standard protocol (IEEE 802.1AB).
- Because they share information about the devices in the network, they can be considered a security risk and are often not used. It is up to the network engineer/admin to decide if they want to use them in the network or not.



It gives the basic concept of CDP/LLDP and shows R1 and SW1 exchanging information.

2. CDP — Cisco Discovery Protocol

CDP = Cisco Discovery Protocol

- Cisco proprietary protocol.
- Used to discover **directly connected Cisco devices**.
- Operates at **Layer 2**.
- Enabled by default on many Cisco devices/interfaces.
- CDP messages are sent periodically.
- CDP messages use multicast MAC address:

0100.0CCC.CCCC

CDP does NOT forward messages

When a device receives a CDP message:

Receive CDP



Process it



Store neighbor information



Do NOT forward it

So CDP information only describes the **direct neighbor**.

Default CDP timers

Setting	Default
CDP advertisement timer	60 seconds
CDP holdtime	180 seconds
CDP version	CDPv2

Holdtime

If R1 receives a CDP advertisement from R2:

R2 → R1

R1 keeps R2's information for the holdtime.

If R1 doesn't receive another CDP message within **180 seconds**, R2 is removed from the CDP neighbor table.

Image placement

Place Image 2



Cisco Discovery Protocol

- CDP is a Cisco proprietary protocol.
- It is enabled on Cisco devices (routers, switches, firewalls, IP phones, etc) by default.
- CDP messages are periodically sent to multicast MAC address 0100.0CCC.CCCC.
- When a device receives a CDP message, it processes and discards the message. It does NOT forward it to other devices.
- By default, CDP messages are sent once every **60 seconds**.
- By default, the CDP holdtime is **180 seconds**. If a message isn't received from a neighbor for 180 seconds, the neighbor is removed from the CDP neighbor table.
- CDPv2 messages are sent by default.



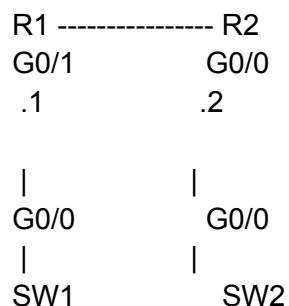
here — CDP

basic characteristics, multicast MAC, 60-second timer, 180-second holdtime.

3. CDP Example

Consider:

10.0.0.0/30



R1 can use CDP to discover:

R1 G0/0 → SW1 G0/0

R1 G0/1 → R2 G0/0

Notice something important:

CDP doesn't need to calculate a route.

It simply answers:

"What device is connected to my interface?"

Image placement



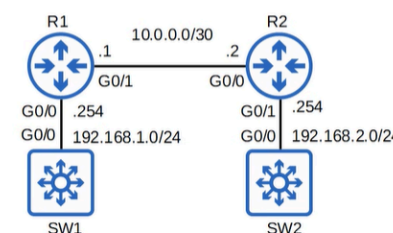
Cisco Discovery Protocol

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled

R1#
R1#show cdp traffic
CDP counters :
  Total packets output: 105, Input: 112
  Hdr syntax: 0, Chksum error: 0, Encaps failed: 0
  No memory: 0, Invalid packet: 0,
  CDP version 1 advertisements output: 0, Input: 0
  CDP version 2 advertisements output: 105, Input: 112

R1#
R1#show cdp interface
GigabitEthernet0/0 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/1 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/2 is administratively down, line protocol is down
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/3 is administratively down, line protocol is down
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds

  cdp enabled interfaces : 4
  interfaces up          : 2
  interfaces down        : 2
```



 **Place Image 3 here** — it shows the topology and CDP global/interface information.

4. show cdp neighbors

The most important CDP command:

show cdp neighbors

It displays basic information about directly connected CDP neighbors.

Example:

Device ID	Local Intf	Holdtime	Capability	Platform	Port ID
SW1	Gig 0/0	153	R S I ...	Gig 0/0	
R2	Gig 0/1	146	R B ...	Gig 0/0	

Important columns

Device ID

→ Name of the neighboring device.

Local Interface

→ Your interface connected to the neighbor.

Holdtime

→ Remaining time before the neighbor is removed if no new advertisement is received.

Capability

→ What the neighbor can act as.

For example:

R = Router

S = Switch

I = IGMP

Platform

→ Device model/platform.

Port ID

→ Interface of the neighboring device.

Image placement

📌 Place Image 4 here — *show cdp neighbors* output



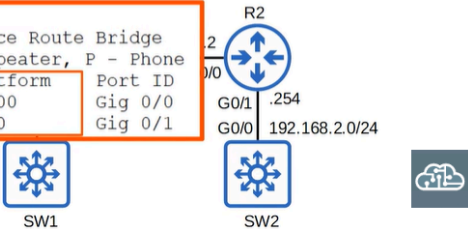
Cisco Discovery Protocol

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
SW1                Gig 0/0         153        R S I
R2                 Gig 0/1         146        R B

Total cdp entries displayed : 2
R1#
```

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
R2                 Gig 0/1         124        R
SW1                Gig 0/0         126        S
                  C2900
                  2960
```



```
graph TD
    SW1[SW1] --- R2[R2]
    SW2[SW2] --- R2
    R2 --- IP[192.168.2.0/24]
```

5. show cdp neighbors detail

For more information:

`show cdp neighbors detail`

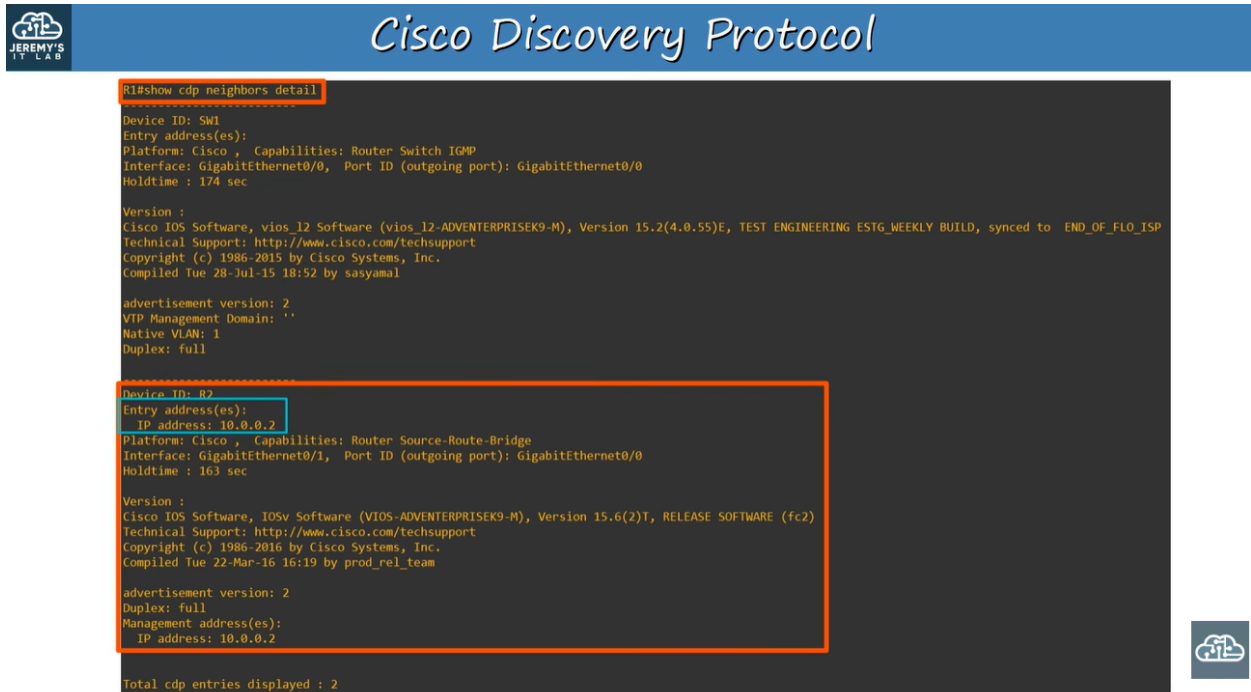
This gives detailed information about each neighbor.

You can see things such as:

- Device ID
- IP address
- Platform
- Capabilities
- Local interface
- Neighbor interface
- IOS version
- Management address
- Holdtime

- CDP version

Image placement



📌 Place Image 5 here — *show cdp neighbors detail* output.

6. show cdp entry

If you only want information about **one specific neighbor**:

show cdp entry R2

Instead of displaying every neighbor, it displays information about R2.

Example:

Device ID: R2

IP address: 10.0.0.2

Platform: Cisco

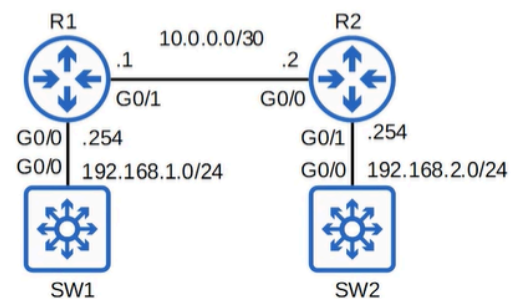
Interface: GigabitEthernet0/1

Port ID: GigabitEthernet0/0

Image placement

```
R1#show cdp entry R2
```

```
-----  
Device ID: R2  
Entry address(es):  
  IP address: 10.0.0.2  
Platform: Cisco , Capabilities: Router Source-Route-Bridge  
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0  
Holdtime : 178 sec  
  
Version :  
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)  
Technical Support: http://www.cisco.com/techsupport  
Copyright (c) 1986-2016 by Cisco Systems, Inc.  
Compiled Tue 22-Mar-16 16:19 by prod_rel_team  
  
advertisement version: 2  
Duplex: full  
Management address(es):  
  IP address: 10.0.0.2
```



📌 Place Image 6 here — *show cdp entry R2* output.

7. CDP Show Commands — Quick Revision

Command	Purpose
<code>show cdp</code>	Basic CDP information
<code>show cdp traffic</code>	CDP packets sent/received
<code>show cdp interface</code>	Interfaces where CDP is enabled

<code>show cdp neighbors</code>	Basic neighbor information
<code>show cdp neighbors detail</code>	Detailed information about neighbors
<code>show cdp entry R2</code>	Detailed information about a specific neighbor

Image placement



CDP show commands summary

- **R1# show cdp**
→ shows basic information about CDP (timers, version)
- **R1# show cdp traffic**
→ displays how many CDP messages have been sent and received
- **R1# show cdp interface**
→ displays which interfaces CDP is enabled on
- **R1# show cdp neighbors**
→ lists CDP neighbors and some basic information about each neighbor
- **R1# show cdp neighbors detail**
→ lists each CDP neighbor with more detailed information
- **R1# show cdp entry name**
→ displays the same info as above, but for the specified neighbor only



 **Place Image 7 here** — *CDP show-command summary.*

8. CDP Configuration

Enable CDP globally

R1(config)# cdp run

Disable CDP globally

R1(config)# no cdp run

Enable CDP on a particular interface

R1(config-if)# cdp enable

Disable CDP on an interface

R1(config-if)# no cdp enable

Change CDP timer

R1(config)# cdp timer 30

Now CDP advertisements are sent every **30 seconds**.

Change CDP holdtime

R1(config)# cdp holdtime 120

Enable/disable CDPv2

R1(config)# cdp advertise-v2

Disable:

R1(config)# no cdp advertise-v2

Image placement



CDP Configuration Commands

- CDP is globally enabled by default.
- CDP is also enabled on each interface by default.
- To enable/disable CDP globally: `R1(config)# [no] cdp run`
- To enable/disable CDP on specific interfaces: `R1(config-if)# [no] cdp enable`
- Configure the CDP timer: `R1(config)# cdp timer seconds`
- Configure the CDP holdtime: `R1(config)# cdp holdtime seconds`
- Enable/disable CDPv2: `R1(config)# [no] cdp advertise-v2`



 Place Image 8 here — CDP configuration commands.

9. LLDP — Link Layer Discovery Protocol

LLDP = Link Layer Discovery Protocol

LLDP performs a job similar to CDP, but there is one major difference:

LLDP is vendor-neutral.

LLDP is an **industry-standard protocol defined by IEEE 802.1AB.**

Therefore:

Cisco ↔ Cisco



CDP or LLDP

Cisco ↔ Aruba



LLDP

Cisco ↔ HP

↓
LLDP

Important

A Cisco device can run **CDP and LLDP at the same time**.

Default LLDP timers

Setting	Default
LLDP advertisement timer	30 seconds
LLDP holdtime	120 seconds
Reinitialization delay	2 seconds

LLDP multicast MAC

0180.C200.000E

Like CDP, LLDP messages are not forwarded through the network.

Image placement

 *Link Layer Discovery Protocol*

- LLDP is an industry standard protocol (IEEE 802.1AB).
- It is usually disabled on Cisco devices by default, so it must be manually enabled.
- A device can run CDP and LLDP at the same time.
- LLDP messages are periodically sent to multicast MAC address 0180.C200.000E.
- When a device receives an LLDP message, it processes and discards the message. It does NOT forward it to other devices.
- By default, LLDP messages are sent once every **30 seconds**.
- By default, the LLDP holdtime is **120 seconds**.
- LLDP has an additional timer called the 'reinitialization delay'. If LLDP is enabled (globally or on an interface), this timer will delay the actual initialization of LLDP. **2 seconds** by default.

 **Place Image 9 here** — LLDP characteristics, IEEE 802.1AB, timers, multicast MAC.

10. LLDP Configuration

Enable LLDP globally

```
R1(config)# lldp run
```

Enable LLDP transmission

```
R1(config-if)# lldp transmit
```

This means:

Send my LLDP information to the neighbor.

Enable LLDP reception

```
R1(config-if)# lldp receive
```

This means:

Receive and learn LLDP information from the neighbor.

Configure LLDP timer

```
R1(config)# lldp timer 30
```

Configure holdtime

```
R1(config)# lldp holdtime 120
```

Configure reinitialization timer

```
R1(config)# lldp reinit 2
```

Image placement



LLDP Configuration Commands

- LLDP is usually globally disabled by default.
- LLDP is also disabled on each interface by default.
- To enable LLDP globally: `R1(config)# lldp run`
- To enable LLDP on specific interfaces (tx): `R1(config-if)# lldp transmit`
- To enable LLDP on specific a interface (rx): `R1(config-if)# lldp receive`
- Configure the LLDP timer: `R1(config)# lldp timer seconds`
- Configure the LLDP holdtime: `R1(config)# lldp holdtime seconds`
- Configure the LLDP reinit timer: `R1(config)# lldp reinit seconds`



 Place Image 10 here — LLDP configuration commands.

11. LLDP Show Commands

The commands are almost identical to CDP.

Basic information

show lldp

Traffic statistics

show lldp traffic

Interface status

show lldp interface

Neighbor information

show lldp neighbors

Detailed neighbor information

show lldp neighbors detail

Specific neighbor

show lldp entry SW1

Image placement

Link Layer Discovery Protocol

```
R1#show lldp traffic

LLDP traffic statistics:
  Total frames out: 4
  Total entries aged: 0
  Total frames in: 3
  Total frames received in error: 0
  Total frames discarded: 0
  Total TLVs discarded: 0
  Total TLVs unrecognized: 0

R1#
R1#show lldp interface

GigabitEthernet0/0:
  Tx: enabled
  Rx: enabled
  Tx state: IDLE
  Rx state: WAIT FOR FRAME

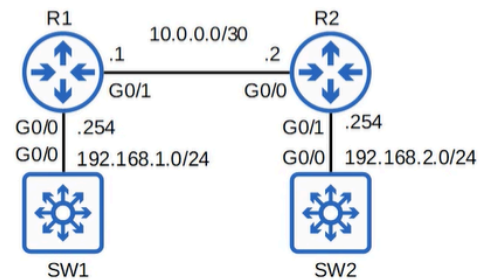
GigabitEthernet0/1:
  Tx: enabled
  Rx: enabled
  Tx state: IDLE
  Rx state: WAIT FOR FRAME

GigabitEthernet0/2:
  Tx: enabled
  Rx: enabled
  Tx state: INIT
  Rx state: WAIT PORT OPER

GigabitEthernet0/3:
  Tx: enabled
  Rx: enabled
  Tx state: INIT
  Rx state: WAIT PORT OPER
```

```
R1#show lldp

Global LLDP Information:
  Status: ACTIVE
  LLDP advertisements are sent every 30 seconds
  LLDP hold time advertised is 120 seconds
  LLDP interface reinitialisation delay is 2 seconds
```



📌 Place Image 11 here — it shows `show lldp traffic`, `show lldp interface`, and `show lldp`.

- R1# **show lldp**
→ shows basic information about LLDP (timers, version)
- R1# **show lldp traffic**
→ displays how many LLDP messages have been sent and received
- R1# **show lldp interface**
→ displays which interfaces LLDP tx/rx is enabled on
- R1# **show lldp neighbors**
→ lists LLDP neighbors and some basic information about each neighbor
- R1# **show lldp neighbors detail**
→ lists each LLDP neighbor with more detailed information
- R1# **show lldp entry name**
→ displays the same info as above, but for the specified neighbor only

 **Place Image 14 here** — LLDP show-command summary.

You can actually **skip Image 14** if you want to avoid repetition, because Image 11 + the command table above already cover it.

12. show lldp neighbors detail

This is the LLDP equivalent of:

show cdp neighbors detail

It can provide:

- Local interface
- Chassis ID
- Port ID
- Port description
- System name
- System description
- Capabilities
- Management address
- Remaining holdtime

Example:

Local Intf: Gi0/0

Chassis id: ...

Port id: Gi0/0

System Name: SW1

System Description:

Cisco IOS Software...

Image placement

Link Layer Discovery Protocol

```
R1#show lldp neighbors detail
-----
Local Intf: Gi0/0
Chassis id: 0c04.41d2.1a00
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: SW1

System Description:
Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERING ESTG_WEEKLY BUILD, synced to END_OF_FLO_ISP
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
Compiled

Time remaining: 99 seconds
System Capabilities: B,R
Enabled Capabilities - not advertised
Management Addresses - not advertised
Auto Negotiation - not supported
Physical media capabilities - not advertised
Media Attachment Unit type - not advertised
Vlan ID: - not advertised

-----
Local Intf: Gi0/1
Chassis id: 0c04.418d.a400
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: R2

System Description:
Cisco IOS Software, IOSv Software (IOSv-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Tue 22-Mar-16 16:19 by prod_rel_team

Time remaining: 92 seconds
System Capabilities: B,R
Enabled Capabilities: R
Management Addresses:
  IP: 10.0.0.2
```

 Place Image 12 here — *show lldp neighbors detail* output.

13. show lldp entry

To view information about one specific LLDP neighbor:

show lldp entry SW1

It gives detailed information about that particular neighbor.

Image placement



 Place Image 13 here — *show lldp entry SW1* output.

14. CDP vs LLDP

This is one of the **most important exam sections**.

Feature	CDP	LLDP
Full name	Cisco Discovery Protocol	Link Layer Discovery Protocol
Standard	Cisco proprietary	IEEE 802.1AB
Vendor	Cisco	Multi-vendor
Layer	Layer 2	Layer 2
Purpose	Neighbor discovery	Neighbor discovery
Cisco ↔ Cisco	✓	✓
Cisco ↔ non-Cisco	✗ Usually not	✓
Default on Cisco	Enabled	Usually disabled

Default timer	60 sec	30 sec
Default holdtime	180 sec	120 sec
Multicast MAC	0100.0CCC.CCCC	0180.C200.000E
Can run simultaneously?		

Memory trick

CDP → Cisco

LLDP → Lots of vendors

15. Security Consideration

This is an important point from your first screenshot.

CDP/LLDP **share information about network devices.**

For example, an attacker who gains access to the network could potentially learn:

- Device name
- IP address
- Interface information
- Device type
- IOS/software information
- Network topology information

Therefore, in security-sensitive environments, administrators may disable CDP/LLDP where they aren't needed.

For example:

```
interface g0/0
no cdp enable
```

The decision depends on the network's security and operational requirements.

Important distinction

This doesn't mean:

"CDP/LLDP are dangerous protocols."

Rather:

They reveal information that may be useful to an attacker, so administrators should decide where they should be enabled.

16. Wireshark — CDP Packet

The CDP Wireshark screenshot shows what is actually inside a CDP frame.

You can see information such as:

Cisco Discovery Protocol

Version: 2

TTL: 180 seconds

Device ID: R1

Software Version

Platform: Cisco

Addresses

Port ID: GigabitEthernet0/0

Capabilities

IP Prefixes

This is useful because it proves that CDP isn't magic.

The device actually sends a **Layer 2 Ethernet frame** containing these pieces of information.

Image placement

📌 Place Image 15 here — Wireshark CDP capture.

Wireshark Captures

```
> Frame 12: 369 bytes on wire (2952 bits), 369 bytes captured (2952 bits) on interface -, id 0
  ✓ IEEE 802.3 Ethernet
    > Destination: CDP/VTP/DTP/PagP/UDLD (01:00:0c:cc:cc:cc)
    > Source: 0c:04:41:47:57:00 (0c:04:41:47:57:00)
    Length: 355
  > Logical-Link Control
    ✓ Cisco Discovery Protocol
      Version: 2
      TTL: 180 seconds
      Checksum: 0xee0f [correct]
      [Checksum Status: Good]
      > Device ID: R1
      > Software Version
      > Platform: Cisco
      > Addresses
      > Port ID: GigabitEthernet0/0
      ✓ Capabilities
        Type: Capabilities (0x0004)
        Length: 8
        ✓ Capabilities: 0x00000005
          ....1 = Router: Yes
          ...0. = Transparent Bridge: No
          ...1.. = Source Route Bridge: Yes
          ...0... = Switch: No
          ...0.... = Host: No
          ...0.... = IGMP capable: No
          ...0... = Repeater: No
          ...0... = VoIP Phone: No
          ...0.... = Remotely Managed Device: No
          ...0.... = CVTA/STP Dispute Resolution/Cisco VT Camera: No
          ...0... = Two Port Mac Relay: No
      > IP Prefixes: 1
      > Duplex: Full
      > Management Addresses
```

17. Wireshark — LLDP Packet

The LLDP Wireshark capture shows:

Link Layer Discovery Protocol

Chassis ID

Port ID

Time To Live

System Name

System Description

Port Description

System Capabilities

You'll also see **TLVs**.

TLV

TLV means:

T → Type

L → Length

V → Value

LLDP uses TLVs to organize information inside its messages.

For example:

Type → System Name

Length → ...

Value → SW1

Another:

Type → Port ID

Length → ...

Value → Gi0/0

Image placement

Wireshark Captures

```
> Frame 466: 325 bytes on wire (2600 bits), 325 bytes captured (2600 bits) on interface -, id 0
> Ethernet II, Src: 0c:04:41:d2:1a:00 (0c:04:41:d2:1a:00), Dst: LLDP_Multicast (01:80:c2:00:00:0e)
▼ Link Layer Discovery Protocol
  > Chassis Subtype = MAC address, Id: 0c:04:41:d2:1a:00
  > Port Subtype = Interface name, Id: Gi0/0
  > Time To Live = 120 sec
  > System Name = SW1
  > [truncated]System Description = Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Versic
  > Port Description = GigabitEthernet0/0
  ▼ Capabilities
    0000 111. .... = TLV Type: System Capabilities (7)
    .... ..0 0000 0100 = TLV Length: 4
    ▼ Capabilities: 0x0014
      .... ..0 = Other: Not capable
      .... ..0. = Repeater: Not capable
      .... ..1.. = Bridge: Capable
      .... ..0... = WLAN access point: Not capable
      .... ..1.... = Router: Capable
      .... ..0. .... = Telephone: Not capable
      .... ..0.. .... = DOCSIS cable device: Not capable
      .... ..0... .... = Station only: Not capable
    ▼ Enabled Capabilities: 0x0010
      .... ..0 = Other: Not capable
      .... ..0. = Repeater: Not capable
      .... ..0.. = Bridge: Not capable
      .... ..0... = WLAN access point: Not capable
      .... ..1.... = Router: Capable
      .... ..0. .... = Telephone: Not capable
      .... ..0.. .... = DOCSIS cable device: Not capable
      .... ..0... .... = Station only: Not capable
  > End of LLDPDU
```

 Place Image 16 here — Wireshark LLDP capture.

18. The Most Important Concept — CDP/LLDP vs OSPF

Since this was your original doubt, keep this in your notes.

CDP / LLDP

Answers:

"Who is directly connected to me?"

Example:

R1 G0/0



SW1 G0/0

CDP/LLDP tells you:

My interface: G0/0

Neighbor: SW1

Neighbor interface: G0/0

OSPF

Answers:

"What networks can I reach, and through which router?"

Example:

```
R1 ----- R2 ----- R3
      |
      192.168.3.0/24
```

OSPF can learn:

192.168.3.0/24

↓

via R2

and install the route into the routing table.

Therefore:

NETWORK

CDP / LLDP

↓

Discover topology

↓

"Who is connected to me?"

+

OSPF

↓

Discover routes

↓

"Where should I send traffic?"

CDP/LLDP helps the network engineer understand the topology.
OSPF helps the router make routing decisions.

Recommended Image Order for Your Notes

If you want your notes to be clean and not overloaded with screenshots, I recommend this exact order:

Order	Image	Where to place
1	Image 1	Layer 2 Discovery Protocol introduction
2	Image 2	CDP characteristics
3	Image 3	CDP example/topology
4	Image 4	<code>show cdp neighbors</code>
5	Image 5	<code>show cdp neighbors detail</code>
6	Image 6	<code>show cdp entry</code>
7	Image 7	CDP show-command summary
8	Image 8	CDP configuration
9	Image 9	LLDP characteristics
10	Image 10	LLDP configuration
11	Image 11	LLDP status/traffic
12	Image 12	<code>show lldp neighbors detail</code>
13	Image 13	<code>show lldp entry</code>
14	Image 14	LLDP show-command summary
15	Image 15	Wireshark CDP
16	Image 16	Wireshark LLDP

★ If you want shorter notes

You can **remove Image 3, Image 7, Image 11, and Image 14**, because they mostly reinforce information already covered.

Then your essential screenshots are:

1 → 2 → 4 → 5 → 6 → 8 → 9 → 10 → 12 → 13 → 15 → 16

That gives you a much cleaner study document without losing any important CCNA concepts.